Voya — Task List, Milestones, Timeline, and Effort Matrix

Assignment #5 and #6 | Team: Tessneem Khalil and Salwa Syed

Planning assumption: Design I (Assignment #4, D0–D2) runs through Fall 2026. The Task List below covers the implementation semester, CS 5002 Senior Design Project II, which we assume runs Monday, January 11 – Friday, April 23, 2027 (14 weeks). If the actual Spring 2027 calendar differs, every date below shifts by the same offset; the durations, dependencies, and buffers do not change.

Task List (Assignment #5)

Sixteen tasks, about eight per member, each sized to one or two weeks and tied to a D0–D2 module.

Tessneem Khalil

1. Specify the C1 Web Client screens and component states for trip entry, search, and the cost-report view. (C1 Web Client) Owner: Tessneem.
2. Develop the C1 Web Client trip-entry and itinerary-draft forms with field validation and idempotent submit. (C1 Web Client) Owner: Tessneem.
3. Develop the C3 Itinerary Service save, edit, and publish endpoints per the D2-3 state machine. (C3 Itinerary Service) Owner: Tessneem.
4. Develop the C3 Itinerary Service remix endpoint with ownership checks and a transactional copy. (C3 Itinerary Service) Owner: Tessneem.
5. Develop the C6 Feedback Console cost-flag submission and admin review endpoints. (C6 Feedback Console) Owner: Tessneem.
6. Develop the C1 Web Client season-search and cost-breakdown display views. (C1 Web Client) Owner: Tessneem.
7. Test concurrency edge cases E-C3-03, E-C3-04, and E-C6-03 (double submit, remix-during-delete, dual reviewer) from the D2 edge-case table. Owner: Tessneem.
8. Validate the C1/C2 interface contract against D0's thirteen interfaces and refine 401/422/503 error handling across every C1 view. Owner: Tessneem.

Salwa Syed

9. Obtain and configure credentials and rate limits for the X1 OIDC provider, X2 climate-data API, and X3 exchange-rate API. (Configuration) Owner: Salwa.
10. Develop the C7 Data Store schema and migrations for User, Destination, Itinerary, Day, CostItem, ClimateNormal, and CostFlag. (C7 Data Store) Owner: Salwa.
11. Develop the C2 API Layer request routing, session authentication, and domain-error-to-status mapping. (C2 API Layer) Owner: Salwa.
12. Develop the Logging glue module (structured stdout logging, request IDs, secret redaction). (Logging) Owner: Salwa.
13. Develop the Auth/Session glue module (OIDC login, callback, logout, signed cookie issuance). (Auth/Session) Owner: Salwa.
14. Develop the C4 Cost Aggregation Engine query and reliability-labeling logic (reliable / limited_data / no_data). (C4 Cost Aggregation Engine) Owner: Salwa.
15. Develop the C5 Season Search Service rating rule and weekly climate-normal refresh job. (C5 Season Search Service) Owner: Salwa.
16. Validate C4/C5 query performance against the D1 100x data-scale estimate using about 2,000 seeded itineraries, closing gap G1. (C4 Cost Aggregation Engine / C5 Season Search Service) Owner: Salwa.

(Saved separately as `Tasklist.md` in the repository root, same numbering and wording.)

Milestones List (Assignment #6)

ID	Milestone	What it demonstrates	Date	Verified by	Owner
M0	Advisor recorded and team contract approved	Dr. Jillian Aurisano confirmed as advisor; contract signed	2026-09 (complete)	Signed contract + spreadsheet entry on Canvas	Both
M1	User Stories, Use Cases, Acceptance Criteria submitted	US-01–US-05, UC-01, AC-01.1/01.2	2026-10-07 (complete)	`User_Stories.md` committed, tagged `week04`	Both
M2	D0 High-Level Design submitted	7 components, 13 interfaces, 2 data-flow diagrams, decision log	2026-10-02 (complete)	PDF + sources in `Design_Diagrams/`	Both
M3	D1 Detailed Design submitted	C3/C4 data model and aggregation algorithm, API contract, tech stack	2026-10-07 (complete)	PDF + sources in `Design_Diagrams/`	Both
M4	D2 Detailed Design + peer review submitted	C1/C2/C5/C6 behavior, state machines, edge-case and traceability tables	2026-10-14 (complete)	PDF + `Peer_Review_Notes.md`	Both
M5	Task List, Milestones, Timeline, Effort Matrix submitted	This document	2026-10-28 (complete on submission)	PDF + `Tasklist.md` committed	Both
M6	Design I Week 14 report / final presentation	Design phase closed out to advisor and instructor	2026-12-02 (planned)	Presentation + report on Canvas	Both
M7	Foundation live in dev: C7 schema, C2 API layer, Configuration, Logging	A deployed dev environment accepts an authenticated request end-to-end	2027-02-11	Live `GET /healthz` call + manual auth smoke test	Salwa
M8	Authenticated itinerary capture working end-to-end	Trip entry → save → edit → publish, C1 through C3 to C7	2027-02-25	Live demo of the save/edit/publish flow (UC against US-02)	Tessneem
M9	Remix and cost intelligence working	Remix, cost breakdown, and season search return real data	2027-03-27	Live demo against UC-01, AC-01.1/01.2, with seeded data	Salwa

M10	Trust and performance validated	Cost flags, concurrency tests, and the G1 performance test all pass	2027-04-12	Passing test suite + a written load-test result for T16	Both
M11	Integration freeze	All sixteen tasks merged and demo script rehearsed	2027-04-18	Full walkthrough of every use case against its acceptance criteria	Both
M12	Final demo / capstone symposium	Live, working Voya demo to advisor and instructor	2027-04-21	Live demo	Both
M13	Design II Week 14 report	Implementation progress reproduces these tables with actuals	2027-04-23	Report submitted on Canvas	Both

Table 1 — Timeline

Each task's completion date already includes a 15–20% schedule buffer. A protected five-day block with no new task starts precedes each major milestone (M7, M9, M11, M12).

Task ID	Statement (short)	Owner	Start	Completion (incl. buffer)	Buffer	Dependencies
T9	Configuration: obtain API credentials	Salwa	2027-01-12	2027-01-15	1 day (~20%)	none
T10	C7 Data Store schema and migrations	Salwa	2027-01-15	2027-01-28	2 days (~15%)	T9
T1	Specify C1 screens and states	Tessneem	2027-01-12	2027-01-23	2 days (~15%)	none
T11	C2 API Layer routing + auth mapping	Salwa	2027-01-28	2027-02-11	2 days (~15%)	T10, T9
T2	C1 trip-entry and draft forms	Tessneem	2027-01-26	2027-02-09	2 days (~15%)	T1, T11 (contract)
—	**Protected block before M7**	Both	2027-02-08	2027-02-11	—	—
M7	**Foundation live in dev**	Both	—	**2027-02-11**	—	T9, T10, T11
T12	Logging glue module	Salwa	2027-02-11	2027-02-13	1 day (~20%)	T11
T13	Auth/Session glue module	Salwa	2027-02-12	2027-02-24	2 days (~15%)	T11
T3	C3 save/edit/publish endpoints	Tessneem	2027-02-10	2027-02-25	3 days (~18%)	T10, T11
M8	**Itinerary capture end-to-end**	Both	—	**2027-02-25**	—	T3, T13
T4	C3 remix endpoint	Tessneem	2027-02-26	2027-03-05	1 day (~15%)	T3
T14	C4 Cost Aggregation Engine logic	Salwa	2027-02-25	2027-03-12	3 days (~18%)	T11, T10
T5	C6 Feedback Console endpoints	Tessneem	2027-03-08	2027-03-19	2 days (~15%)	T10, T11
T15	C5 Season Search Service + refresh job	Salwa	2027-03-15	2027-03-27	3 days (~20%, spans spring break)	T11, T10
—	**Protected block before M9**	Both	2027-03-24	2027-03-27	—	—
M9	**Remix and cost intelligence working**	Both	—	**2027-03-27**	—	T4, T14, T15
T6	C1 season-search and cost views	Tessneem	2027-03-22	2027-04-02	2 days (~15%)	T14, T15
T16	C4/C5 performance validation (closes G1)	Salwa	2027-03-29	2027-04-09	2 days (~15%)	T14, T15, T10
T7	Concurrency tests E-C3-03/04, E-C6-03	Tessneem	2027-04-05	2027-04-12	1 day (~15%)	T4, T5
M10	**Trust and performance validated**	Both	—	**2027-04-12**	—	T7, T16
T8	C1/C2 interface + error-handling validation	Tessneem	2027-04-12	2027-04-16	1 day (~15%)	T6, T7
—	**Protected block before M11/M12**	Both	2027-04-16	2027-04-20	—	—
M11	**Integration freeze**	Both	—	**2027-04-18**	—	T8, T16
M12	**Final demo / symposium**	Both	—	**2027-04-21**	—	M11
M13	**Design II Week 14 report**	Both	—	**2027-04-23**	—	M12

Table 2 — Effort Matrix

Hours estimated with PERT: (Optimistic + 4×Most-likely + Pessimistic) / 6. Each task's primary owner carries the larger share; the partner's hours cover code review and integration testing on that task.

Task	Tessneem (hrs)	Salwa (hrs)	Total (hrs)	Primary owner
T1	10	2	12	Tessneem
T2	16	3	19	Tessneem
T3	17	3	20	Tessneem
T4	10	2	12	Tessneem
T5	12	2	14	Tessneem
T6	14	3	17	Tessneem
T7	10	2	12	Tessneem
T8	9	2	11	Tessneem
T9	1	6	7	Salwa
T10	2	12	14	Salwa
T11	3	16	19	Salwa
T12	1	6	7	Salwa
T13	2	12	14	Salwa
T14	3	16	19	Salwa
T15	3	14	17	Salwa
T16	2	10	12	Salwa
Column total	**115**	**111**	**226**	—

Balance note: the columns are within 4 hours of each other (115 vs. 111), so no rebalancing is planned. If either partner's co-op rotation or exam schedule removes them for more than a week (see risk R3), we will shift a same-owner task into the other partner's queue using the support hours already budgeted above, rather than rewriting the matrix.

Work Breakdown Structure

- **Phase 1 — Core Services Foundation**
  - T9 Configuration
  - T10 C7 Data Store
  - T11 C2 API Layer
  - T12 Logging
  - T13 Auth/Session
- **Phase 2 — Trip Capture and Itinerary Lifecycle**
  - T1 C1 screen specification
  - T2 C1 trip-entry forms
  - T3 C3 save/edit/publish
  - T4 C3 remix
- **Phase 3 — Discovery and Cost Intelligence**
  - T14 C4 Cost Aggregation Engine
  - T15 C5 Season Search Service
  - T6 C1 season-search and cost views
- **Phase 4 — Trust, Quality, and Demo Readiness**
  - T5 C6 Feedback Console
  - T7 Concurrency tests
  - T16 Performance validation (G1)
  - T8 Interface and error-handling validation

Risk Register

ID	Category	Risk	Likelihood	Impact	Strategy	Mitigation	Owner	Trigger
R1	Technical	Few real users means most destinations sit in `no_data` state, so the demo looks empty	High	Medium	Reduce	Seed ~2,000 synthetic itineraries by T16 (Week 11) so demo destinations are always `reliable`	Salwa	Fewer than 3 itineraries exist for a planned demo destination two weeks before M12
R2	Technical	X2 (climate) or X3 (exchange-rate) API changes schema or rate limits, breaking C5/C3	Medium	Medium	Reduce	Cache normals/rates in C7 (D04/D07) with a stored-fallback path so provider downtime degrades to stale data instead of failure	Salwa	Provider returns non-200 in dev/CI for more than 24 hours
R3	Team	A GE Aerospace co-op rotation, exam week, or travel removes one partner for 1–2 weeks	Medium	High	Reduce	Support hours in Table 2 keep the non-owner partner current enough to cover a blocked task for up to a week; flag any known conflict to the advisor 2+ weeks ahead	Both	Either partner announces a conflict or overlapping a task's planned window

R4	External	UC spring break falls inside the T15/T5 window (mid-March) and reduces available hours that week	High	Low–Medium	Accept, with buffer	The 15–20% task buffer absorbs a short low-output week; no new task start is scheduled during the break itself	Both	Calendar-certain; reviewed at the M9 check-in
R5	Technical	The 3-second AC-01.1 performance target (open gap G1) fails once C4 runs against realistic data on a free-tier host	Medium	High	Reduce	T16 runs a dedicated load test against ~2,000 seeded itineraries with 3 weeks of buffer before M11 to add an index or caching if missed	Salwa	T16's measured p95 response time exceeds 3 seconds
R6	Team	Scope creep adds an out-of-scope feature (e.g., full moderation, review voting — D2 gap G5) that consumes budgeted time	Medium	Medium	Avoid	Any new feature is checked against the D2 Requirements Traceability Table before being added; untraceable features go to a Phase-2 backlog instead of this plan	Both	A feature is proposed with no US-xx or UC-xx ID behind it
R7	External	The free/low-cost hosting tier sleeps or rate-limits an idle service, causing a cold-start delay right before a graded demo	Medium	Medium	Reduce	Confirm sleep/idle behavior during T11 (Week 4); add a keep-alive ping or upgrade tier before M7 if the service sleeps	Salwa	Dev deployment cold-starts (>5s to first response) after idle
R8	Technical	Stdout-only logging (accepted in D2, E-LG-04) means log data is lost if the host's log pipeline drops output, hiding a late bug's cause	Low	Low	Accept	None beyond the existing design — `GET /healthz` and the database remain the source of truth for debugging	Salwa	Accepted; revisited only if logs are needed for a specific incident

High-impact schedule risks (R1, R3, R5, R7) are reflected in Table 1: R1 and R5 are why T16 sits three weeks before the integration freeze (M11) rather than immediately before it, R3 is why every task carries partner support hours in Table 2, and R7 is why hosting behavior is checked during T11 in Week 4 rather than discovered at demo time.